

Measured Dynamical Extent

From Substance to Stability

Kevin R. Haylett
Manchester, UK

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Pensée

Abstract

The physical sciences have inherited a language in which *matter* is commonly treated as a primitive: matter exists, possesses properties, occupies space, and subsequently undergoes change. This paper considers an alternative starting point drawn from Finite Symbolic Mechanics (FSM) and Finite Symbolic Dynamics (FSD). Rather than beginning with matter as substance, it proposes the concept of *Measured Dynamical Extent* (MDE): a finite registered extent within which relational dynamics exhibit distinguishable patterns of stability.

Within this formulation, familiar physical categories such as solid, liquid, gas, plasma, condensate and vacuum are not initially treated as different kinds of substance. They are treated as symbolic descriptions of different regimes of measured dynamical stability. A physical “thing” is consequently not required as a primitive. It may instead arise as a compressed symbolic designation for a sufficiently persistent pattern of registered relations.

This shift also requires a refinement of the Alphonic Limit. The Alphonic Limit cannot consistently be interpreted as a smallest static object or geometrical bead. If all admissible physical distinction requires finite registration, then the smallest distinction must itself possess dynamical and temporal extent. The Alphonic Limit is therefore reconsidered as the minimum finite dynamical extent capable of sustaining a distinguishable registration.

The purpose of the paper is deliberately limited. It does not attempt to redefine mass, gravity, charge, quantum theory or cosmology. It establishes a conceptual and operational foundation from which those questions may subsequently be investigated without beginning from an assumed ontology of substance.

Keywords: Finite Symbolic Mechanics; Finite Symbolic Dynamics; measured extent; stability; Alphonic Limit; registration; phase; dynamical systems; symbolic compression.

1 Introduction

Physics is commonly narrated through nouns.

There are particles, bodies, fields, atoms, molecules, solids, fluids, gases, plasmas and vacua. These entities are then assigned properties and equations describing what they do.

The grammatical ordering is easily overlooked:

$$\text{thing} \longrightarrow \text{properties} \longrightarrow \text{behaviour.} \quad (1)$$

The thing arrives first.

Its dynamics arrive afterwards.

This ordering is so deeply embedded in scientific language that it can appear unavoidable. Yet measurement itself does not necessarily present the observer with a pre-labelled thing.

Measurement presents finite registrations: changes of detector state, marks, counts, voltages, displacements, frequencies, images, timings and other distinguishable outputs.

The statement that these registrations belong to a particular “thing” is already an act of organisation.

Finite Symbolic Mechanics begins from this finite act of distinction rather than from an assumed infinitely precise world of mathematical objects. Finite Symbolic Dynamics extends this concern into changing systems, where what is registered is not merely a value but an ordered trajectory of finite symbolic states.

The present paper asks whether the inherited physical primitive of *matter* is necessary within such a framework.

The proposed replacement is:

Measured Dynamical Extent.

This expression is not intended merely as new terminology for an old object. Its purpose is to reverse the direction of description.

Instead of:

$$\text{matter} \longrightarrow \text{state} \longrightarrow \text{dynamics}, \quad (2)$$

we begin with:

$$\text{extent} \longrightarrow \text{interaction} \longrightarrow \text{dynamics} \longrightarrow \text{stability} \longrightarrow \text{registration} \longrightarrow \text{symbolic state}. \quad (3)$$

The distinction is substantial.

Matter begins by asserting a category.

Measured Dynamical Extent begins by asking what can be distinguished.

2 The Difficulty with Matter

The word *matter* performs an enormous amount of conceptual work.

A rock is matter.

Water is matter.

Air is matter.

Plasma is matter.

A crystal is matter.

Yet these categories exhibit radically different dynamical behaviour. The unity provided by the word matter is therefore not directly a unity of observed behaviour. It is a higher-level classification imposed across many different regimes.

This classification may be useful. Usefulness, however, does not require ontological priority.

Within FSM, a distinction is made between what is registered and what is subsequently symbolised. A symbol is finite, portable and compressive. It allows a long and complicated trajectory of registration to be replaced by a comparatively small representational token.

The word

ROCK

does not contain the trajectory by which a particular rock was encountered, measured, weighed, photographed, struck, heated, chemically examined or otherwise distinguished.

It compresses that history.

Likewise,

MATTER

compresses an even larger set of distinctions.

The danger occurs when the compressed term is subsequently treated as though it had preceded the observations from which it was formed.

In symbolic form:

$$\text{registered trajectory} \longrightarrow \text{symbol} \tag{4}$$

may silently become interpreted as:

$$\text{symbolic entity} \longrightarrow \text{cause of registered trajectory.} \tag{5}$$

The map has been allowed to become the source of the territory it describes.

Measured Dynamical Extent is intended to interrupt this reversal.

3 Extent Before Substance

The word *extent* is deliberately weaker than the word *matter*.

It does not initially tell us what occupies the extent.

It does not require atoms.

It does not require particles.

It does not require a field.

It does not require a continuum.

It states only that some finite distinction possesses measurable spread or persistence sufficient to participate in registration.

Extent also prevents the geometrical point from entering unnoticed as a primitive physical object. A zero-extent point may function within a formal mathematical language, but no physical registration is itself without extent.

Every mark has extent.

Every detector has extent.

Every interaction has duration.

Every measured difference occupies some finite interval of distinguishability.

Accordingly, physical description begins not with the dimensionless point but with finite distinction.

The qualification *dynamical* adds a second requirement.

The measured extent is not assumed to be frozen.

Its relations may persist, oscillate, fluctuate, reorganise, bifurcate, decay or stabilise. Even what appears static at one observational scale may exhibit extensive dynamics at another.

A crystal resting on a laboratory bench is macroscopically stable. At another scale and resolution it exhibits thermal motion, lattice vibration, defects, local rearrangements and other dynamics.

The statement

“the crystal is stationary”

is therefore conditional.
It means approximately:

“no change relevant to the present measurement has exceeded the operative threshold of distinction during the specified interval.”

Static description becomes a limiting case of measured dynamical description.

4 Measured Dynamical Extent

We may now give a preliminary definition.

Measured Dynamical Extent (MDE): A finite registered extent within which relational dynamics exhibit a distinguishable pattern of stability under specified conditions of spatial resolution, temporal resolution, interaction, uncertainty and provenance.

The definition is intentionally operational without being ontologically exhaustive.

It does not claim that an unmeasured world does not exist.

It states instead that FSM makes no physical-symbolic claim beyond what can enter a finite trajectory of registration.

This distinction is essential.

There may be exogenous dynamics that have not been measured. Indeed, measurement frequently begins precisely because such dynamics are suspected.

The framework therefore allows a distinction between:

$$\mathcal{E}_D \tag{6}$$

for inferred dynamic extent, and

$$\mathcal{E}_M \tag{7}$$

for measured dynamical extent.

The relationship may be represented schematically as

$$\mathcal{E}_D \xrightarrow{\mathcal{G}} \mathcal{E}_M \longrightarrow S, \tag{8}$$

where $\mathcal{G}$ denotes a generonic interaction or registration boundary and S denotes the resulting symbolic state.

The first term remains inferred.

The second is registered.

The third is compressed.

This gives three different epistemic conditions that should not be silently merged.

5 The Measurement Conditions

A measured dynamical extent cannot be specified independently of the conditions under which it is distinguished.

A useful preliminary representation is

$$\mathcal{E}_M = \mathcal{E} \mid (\Delta x, \Delta t, I, \alpha, \delta, H), \quad (9)$$

where:

- Δx represents operative spatial resolution;
- Δt represents operative temporal resolution;
- I represents the relevant interaction or measurement relation;
- α represents the operative Alphonic condition;
- δ represents retained uncertainty;
- H represents provenance or measurement history.

This notation should not be mistaken for a claim that these variables exhaust the physical situation.

Its purpose is to preserve what conventional compressed notation frequently removes.

A statement about extent is always made under finite conditions.

The state registered at one spatial or temporal resolution need not be the state registered at another.

Consequently,

$$S = S(\Delta x, \Delta t, I, \alpha, \delta, H). \quad (10)$$

This means that physical state is relationally and instrumentally situated.

It does not mean that the state is arbitrary.

Different observers using compatible instruments, procedures and admissibility conditions may obtain highly reproducible registrations. Consensus may therefore become strong.

But reproducibility does not remove the measurement conditions under which reproduction occurs.

6 Phases as Stability Regimes

The concept of physical phase provides an immediate application.

Solid, liquid, gas, plasma, condensate and related classifications are traditionally described as states or phases of matter.

MDE allows a different formulation:

A physical phase is a recurrently distinguishable regime of stability within measured dynamical extent.

This formulation moves the emphasis away from substance and towards behaviour.

A crystalline phase may be characterised by persistent relational organisation over specified scales.

A fluid phase may retain local interaction while permitting continual large-scale rearrangement.

A gaseous phase may exhibit another characteristic range of persistence, separation and interaction.

A plasma may display another relational regime.

A condensate may exhibit unusually extended coherence.

The crucial point is that none of these descriptions requires the phase name to be treated as an irreducible kind of thing.

Instead:

$$\text{dynamical organisation} \longrightarrow \text{persistent regime} \longrightarrow \text{registered distinction} \longrightarrow \text{phase name.} \quad (11)$$

The name is downstream of the stability.

7 A Phase Diagram Reconsidered

Under this interpretation, a phase diagram can be reconsidered as a map of stability regimes.

Its axes represent measured variables or controlled conditions.

Its regions represent domains over which recognisably persistent forms of organisation are registered.

Its boundaries represent transitions between such organisations.

The conventional interpretation may be stated as:

$$\text{conditions} \longrightarrow \text{which phase of matter exists?} \quad (12)$$

The MDE interpretation becomes:

$$\text{conditions} \longrightarrow \text{which dynamical organisation remains stable?} \quad (13)$$

This subtle shift changes the significance of the boundary.

A phase boundary is no longer merely the symbolic dividing line between two named kinds of material state.

It identifies the region in which one pattern of stability ceases to remain admissibly persistent and another emerges.

Schematically:

$$S_1 \longrightarrow B \longrightarrow S_2, \quad (14)$$

where S_1 and S_2 are registered stability regimes and B denotes a transition region.

From the perspective of nonlinear dynamics, such regions may involve bifurcation, metastability, hysteresis, critical behaviour, competing attractors or other reorganisations.

MDE does not prescribe which dynamical model must apply.

It merely relocates the conceptual emphasis from a changing substance to a changing stability regime.

8 Stability Does Not Mean Stillness

The word *stability* requires care.

A stable dynamical system need not be motionless.

Oscillation may be stable.

Rotation may be stable.

A chaotic attractor may possess stable statistical structure.

A turbulent flow may retain persistent large-scale organisation while undergoing rapid local change.

A living organism may retain recognisable identity while continuously exchanging material and reorganising internally.

Therefore the phrase *pattern of stability* is preferable to an everyday notion of stability as absence of change.

A measured state may remain distinguishable precisely through its characteristic mode of change.

In this sense:

$$\text{stability} \neq \text{stasis}. \quad (15)$$

Rather,

$$\text{stability} \sim \text{persistence of distinguishable relational organisation}. \quad (16)$$

The symbol $\sim$ is used deliberately here to indicate a relation rather than an assertion of exact identity.

This distinction considerably broadens the scope of the concept.

Solids, fluids, gases, plasmas, oscillators, turbulent systems and biological structures can all be described within the same general language without claiming that their forms of stability are equivalent.

9 Scale and the Production of State

A central consequence follows.

The apparent state of an extent depends upon the scale over which its stability is registered.

Consider again a crystal.

At human spatial and temporal scales it may appear rigid.

At smaller scales its internal organisation is dynamic.

At still finer scales additional dynamics may become distinguishable.

There is therefore no scale-independent symbolic description called simply “the state of the crystal”.

There are registered states under declared conditions.

The same principle applies temporally.

A structure that is stable over one millisecond may not be stable over one hour.

A structure stable for one year may not remain so over geological durations.

The compression

$$\text{SOLID} \quad (17)$$

therefore suppresses a large quantity of temporal, spatial and instrumental qualification.

This suppression is frequently useful.

But FSM requires us to remember that the qualification has been compressed rather than eliminated.

A symbolic state is consequently better understood as a finite statement about persistence across an admissible observational window.

10 Vacuum as a Measured Stability Regime

Vacuum provides an especially revealing case.

The ordinary word encourages an immediate interpretation:

$$\text{vacuum} = \text{nothing.} \quad (18)$$

But an experimental vacuum is not produced by measuring nothing.

It is produced through apparatus, pumping, containment, pressure measurement, temperature control, electromagnetic conditions, residual-gas estimation, detector thresholds and finite observational intervals.

The word *vacuum* therefore compresses an extensive experimental trajectory.

MDE permits a deliberately restrained interpretation:

Vacuum is a named measured stability regime of dynamical extent.

No further ontological claim is required.

This definition does not assert what vacuum ultimately consists of.

It does not require the introduction of virtual particles, fields, hidden media or a null substrate.

It simply recognises that a region described experimentally as vacuum is still a finite measured region displaying a reproducible pattern of registrations.

Thus:

$$\text{VACUUM} \quad (19)$$

is not synonymous with an experimentally established “nothing”.

It is a symbolic classification of a particular measured regime.

This distinction may be small linguistically, but it is foundationally important.

11 The Alphonic Limit Must Be Dynamical

The introduction of MDE requires a reconsideration of the Alphonic Limit.

Within FSM, the Alphonic Limit prevents the physical point of zero extent from entering measurement unnoticed.

The smallest admissible distinction cannot have zero spatial extent.

However, if measurement is dynamical, then spatial extent alone is insufficient.

No registration occurs at literally zero duration.

A detector changes state across an interval.

A distinction persists sufficiently to affect another system.

A mark remains distinguishable.

A signal crosses a threshold.

A registration is therefore not merely spatial.

It possesses temporal extension.

This means that the Alphonic Limit cannot consistently be imagined as the smallest static sphere or bead from which the world is assembled.

Such an interpretation would simply recreate atomism at a finer descriptive level.

Instead, the Alphonic Limit should be understood dynamically.

A preliminary definition is:

Alphonic Dynamical Extent: The minimum finite dynamical extent over which a relational difference can remain sufficiently distinguishable to participate in an admissible registration.

This changes the foundational question.
Rather than asking:

What is the smallest thing that exists?

or even:

What is the smallest volume that can be measured?

we ask:

What is the minimum dynamical extent capable of becoming a stable finite distinction?

This is a significantly different question.

12 From Resolution Limit to Stability Limit

The Alphonic Limit has often been approached naturally as a resolution limit.

MDE suggests that it may also need to be understood as a stability limit.

A distinction must persist sufficiently to cross a measurement boundary.

Below some operative condition, the difficulty may not be that “nothing exists”.

The difficulty may instead be that no relational difference remains sufficiently stable, relative to the interaction and instrument, to become a finite registered symbol.

This produces a sequence:

$$\text{difference} + \text{extent} + \text{persistence} + \text{interaction} \longrightarrow \text{registration.} \quad (20)$$

Only then can:

$$\text{registration} \longrightarrow \text{symbol.} \quad (21)$$

Thus the Alphonic Limit does not designate a hidden ultimate object.

It designates the operational boundary at which distinction becomes symbolically admissible.

The precise value of such a limit need not be universal across every measurement domain.

Different instruments, interactions and physical processes may possess different operative limits.

The foundational commitment is therefore not to a particular smallest object, but to finite distinction itself.

13 The Generonic Boundary

The generonic boundary becomes particularly important under this interpretation.

The term denotes the transition at which an analogue, exogenous or continuously changing interaction becomes a finite registered distinction.

The boundary is not necessarily a geometrical surface.

It may be implemented physically by a detector threshold, analogue-to-digital conversion, chemical transition, mechanical displacement, neuronal response or other finite state change.
Schematically:

$$\text{dynamic interaction} \xrightarrow{\mathcal{G}} \text{registered distinction.} \quad (22)$$

The generonic operation therefore converts dynamical extent into symbolic availability.

This makes the measurement chain central to physical description.

The symbolic output does not contain the entire preceding dynamics.

It is a compression.

A detector value, for example, may represent an extensive interaction history while carrying only a small finite symbolic result.

FSM therefore distinguishes between the trajectory and its terminal symbol.

MDE belongs primarily to the trajectory side of this boundary.

Named physical entities belong increasingly to the symbolic side.

14 The Emergence of the Thing

The reversal introduced by MDE can now be stated clearly.

The classical descriptive sequence is approximately:

$$\boxed{\text{Thing} \rightarrow \text{Properties} \rightarrow \text{Behaviour}} \quad (23)$$

The proposed sequence is:

$$\boxed{\text{Dynamical Extent} \rightarrow \text{Interaction} \rightarrow \text{Stability} \rightarrow \text{Distinction} \rightarrow \text{Registration} \rightarrow \text{Name}} \quad (24)$$

The thing arrives at the end.

A rock may therefore be interpreted not initially as a primitive object possessing stability, but as a symbolic compression applied to a region of dynamical extent whose relational organisation persists sufficiently across the relevant observational scales.

Likewise, the categories crystal, droplet, gas cloud and plasma may all be regarded as terminal symbolic compressions of different histories of registration.

This does not make the categories unreal.

Their utility and reproducibility may be extremely high.

It merely relocates their status.

They are not required as primitive constituents of the description.

They are stable symbolic achievements.

15 Nouns as Stability Compression

This physical argument connects directly to a broader linguistic problem.

A noun frequently presents a process as though it were a thing.

The word *river* appears stable while the water changes continuously.

The word *flame* names a persistent dynamical organisation despite rapid replacement of participating material.

The word *organism* maintains symbolic identity across metabolism, growth, repair and exchange.

The word *storm* compresses a vast moving atmospheric trajectory into a portable nominal unit.

Physical nouns may behave similarly.

A noun can therefore be understood as a powerful device for compressing persistence.

This suggests the general trajectory:

$$\text{dynamic relations} \longrightarrow \text{recurrent stability} \longrightarrow \text{symbolic compression} \longrightarrow \text{noun}. \quad (25)$$

The noun is useful precisely because the underlying trajectory does not need to be reconstructed every time the term is used.

But the success of compression can conceal its own origin.

Once established, the noun may appear more fundamental than the dynamics from which it arose.

MDE attempts to restore the trajectory beneath the noun.

16 Relation to Finite Symbolic Dynamics

Finite Symbolic Dynamics concerns the evolution of finitely registered symbolic states.

If MDE describes the measured dynamical extent from which state distinctions arise, then FSD describes the ordered transitions among those distinctions.

Let a sequence of registered states be

$$S_0 \mid S_1 \mid S_2 \mid \cdots \mid S_n. \quad (26)$$

This sequence is not the physical dynamics themselves.

It is their finite symbolic trajectory.

The distinction is crucial.

The continuous or exogenous interaction history may contain changes that are not preserved in the registered sequence.

The FSD trajectory therefore represents a measured projection or compression of the dynamical extent.

This can be written schematically as

$$\mathcal{E}_D(t) \xrightarrow{\mathcal{G}} \{S_0, S_1, \dots, S_n\}. \quad (27)$$

The resulting trajectory retains order and provenance while remaining finite.

This gives MDE and FSD complementary roles.

Measured Dynamical Extent concerns the finite registered physical regime.

Finite Symbolic Dynamics concerns the ordered symbolic evolution of the registrations generated from that regime.

17 Relation to Functional Symbolic Trajectories

The same operation continues into language and reasoning.

A registered physical distinction may become a symbol.

Symbols may be linked into ordered trajectories.

Those trajectories may subsequently become concepts, descriptions, equations, models and theories.

A possible chain is:

dynamic extent $\rightarrow$ registration $\rightarrow$ symbol $\rightarrow$ symbolic trajectory $\rightarrow$ compressed concept. (28)

The Functional Symbolic Trajectory (FST) therefore extends the same structural concern beyond the immediate physical measurement.

The word *matter*, for example, is not merely a dictionary entry.

It carries a historical trajectory of prior distinctions, uses, expectations, equations, experiments and conceptual commitments.

When the word enters an argument, much of that trajectory remains hidden.

Replacing *matter* with *Measured Dynamical Extent* is therefore not simply terminological substitution.

It changes the trajectory available to subsequent reasoning.

The phrase makes explicit what the noun suppresses:

- that the description is measured;
- that the described extent is dynamical;
- that stability is conditional rather than absolute;
- that the physical state is finitely registered;
- that the resulting category is symbolic.

The expanded phrase carries more trajectory and therefore produces less immediate symbolic compression.

That is precisely its purpose.

18 Interactional Density

The concept of interactional density may offer one means of characterising different regimes of MDE.

A preliminary expression might be written:

$$\rho_I \sim \frac{\text{registered interactional change}}{\text{measured dynamical extent}}. \quad (29)$$

This should not yet be treated as a completed physical quantity.

Its purpose at this stage is conceptual.

Different dynamical regimes may be distinguishable not simply through conventional measures such as mass density, temperature or pressure, but through the organisation, persistence and concentration of interactions across an extent.

A crystalline state and gaseous state may differ not because one contains “more matter” in any fundamental sense, but because the spatial and temporal relations between registrations occupy different stability regimes.

Interactional density may eventually provide one descriptor within a larger finite state vector:

$$\mathbf{D}_E = (\rho_I, \tau_S, \lambda_R, C, \delta, \dots), \quad (30)$$

where τ_S might represent a measured persistence interval, λ_R a relational scale, and C some admissible measure of coupling or coherence.

No claim is made here that these are the correct or complete variables.

The suggestion is simply that the classification of physical phases may ultimately be approached through descriptors of dynamical stability rather than through a prior ontology of substance.

19 Measured Extent and Continuum

MDE also places a constraint on the use of continuum descriptions.

A continuum may remain an exceptionally useful mathematical construction.

The concern arises only when the continuous symbolic representation is silently promoted into an exact physical statement.

Every actual registration remains finite.

Every numerical result possesses finite symbolic extent.

Every instrument has finite resolution.

Every measurement has temporal duration.

The continuum therefore belongs to the representational language rather than automatically to the registered world.

MDE preserves this distinction by requiring a physical extent to enter the theory through finite measurement.

One may model that extent continuously where useful, but the model should not erase the conditions through which the extent became symbolically available.

Thus:

$$\text{finite registration} \longrightarrow \text{continuous model} \quad (31)$$

does not licence the reverse inference:

$$\text{continuous model} \therefore \text{physically continuous ontology.} \quad (32)$$

The latter remains an additional commitment.

20 What MDE Does Not Yet Claim

A foundational concept is strengthened by making its limits explicit.

Measured Dynamical Extent does not, by itself, establish a new theory of gravity.

It does not establish that mass is attractor curvature.

It does not establish that inertia is symbolic drag.

It does not establish that dark matter is unmeasured extent.

It does not establish a mechanism for charge.

It does not reconstruct quantum mechanics.

It does not identify the vacuum with any particular hidden dynamical substrate.

Each of these may become a legitimate direction of investigation.

None follows automatically from the definition of MDE.

The purpose of the present proposal is prior to those questions.

It asks whether physical theory can begin without assuming matter as an ontological primitive.

If the answer is yes, subsequent physical concepts may then be reconsidered individually within the language of finite registered dynamics.

This retains uncertainty.

It also prevents the new terminology from becoming a replacement metaphysics.

21 A Minimal Set of Propositions

The central argument can be condensed into a small number of propositions.

Proposition 1: Physical registration is finite

Every experimentally available physical distinction enters symbolic description through a finite interaction occurring over finite spatial and temporal extent.

Proposition 2: Registered physical states are dynamical

A registered state does not imply absolute stillness. It denotes a distinguishable pattern of relational persistence across specified observational conditions.

Proposition 3: Physical phases are stability regimes

Terms such as solid, liquid, gas, plasma, condensate and vacuum may be treated as symbolic classifications of recurrent regimes of measured dynamical stability rather than as primitive kinds of substance.

Proposition 4: State is scale-conditioned

A symbolic physical state is conditional upon the operative spatial resolution, temporal interval, interaction, uncertainty, provenance and Alphonic boundary.

Proposition 5: The Alphonic Limit is dynamical

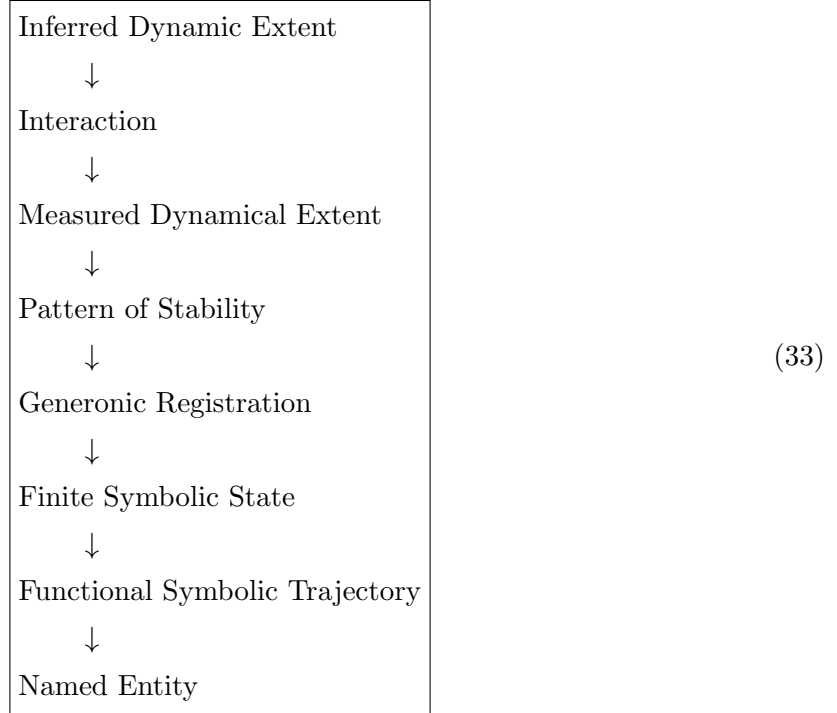
The minimum admissible physical distinction cannot possess zero duration. The Alphonic Limit must therefore be interpreted as a minimum dynamical extent capable of sustaining registration, not as a smallest static thing.

Proposition 6: The entity is downstream of stability

A physical “thing” may be understood as a symbolic compression applied to sufficiently persistent registered relational dynamics.

22 An Emerging Hierarchy

The resulting framework may be represented as:



This hierarchy should not be mistaken for a claim that nature itself proceeds through these linguistic stages.

It is a description of the route by which an exogenous physical interaction becomes available to finite symbolic reasoning.

The importance of the hierarchy lies precisely in keeping these stages apart.

The measurement is not the symbol.

The symbol is not the trajectory.

The trajectory is not the inferred exogenous dynamics.

The name is not necessarily the thing-in-itself.

23 From Thing to Stability

The conceptual reversal may now be stated in its shortest form.

Classical substance-oriented description tends towards:

$$\boxed{\text{Thing} \rightarrow \text{Behaviour}} \quad (34)$$

Measured Dynamical Extent proposes:

$$\boxed{\text{Dynamics} \rightarrow \text{Stability} \rightarrow \text{Distinction}} \quad (35)$$

Finite Symbolic Mechanics then adds:

$$\boxed{\text{Distinction} \rightarrow \text{Registration} \rightarrow \text{Symbol}} \quad (36)$$

and Functional Symbolic Trajectories continue:

$$\boxed{\text{Symbol} \rightarrow \text{Trajectory} \rightarrow \text{Name.}} \quad (37)$$

The physical noun has moved from the beginning of the description to its end.

This does not abolish nouns.

It explains their extraordinary usefulness.

A noun allows a persistent dynamical regime to be carried forward without reproducing the complete trajectory from which it was derived.

The cost is compression.

The scientific problem begins when the compression is forgotten.

24 Open Questions

Measured Dynamical Extent establishes a foundation rather than a completed mechanics.

Several questions immediately follow.

First, what descriptors are sufficient to distinguish stability regimes without simply recreating conventional material properties under new names?

Second, can an operational measure of interactional density be constructed that remains finite and instrumentally admissible?

Third, should the Alphonic Dynamical Extent be treated as a single coupled spatial–temporal condition rather than separated into spatial and temporal limits?

Fourth, how should transitions between stability regimes be represented within Finite Symbolic Dynamics?

Fifth, can classical phase diagrams be reconstructed directly as finite maps of dynamical stability basins?

Sixth, what relationship exists between persistence, registration and the emergence of apparently localised entities?

Seventh, which conventional physical quantities remain useful unchanged within MDE, which require reinterpretation, and which depend upon assumptions that no longer remain admissible?

These are questions for subsequent work.

The present proposal should not prejudge their answers.

25 Conclusion

Measured Dynamical Extent offers a modest but foundational reversal.

Instead of assuming that the physical world is fundamentally composed of things called matter which subsequently possess dynamics, the framework begins with finite measurable extent exhibiting relational change.

Within this extent, patterns of stability emerge.

Those patterns may persist.

Persistent patterns may become distinguishable.

Distinctions may cross generonic boundaries into finite registrations.

Registrations may become symbols.

Symbols may become names.

The familiar “thing” consequently appears at the end of a trajectory of stabilisation and compression rather than at its beginning.

This change is especially useful when considering physical phases.

A crystal, fluid, gas, plasma, condensate or vacuum need not initially be understood as a different ontological substance. Each may instead be approached as a recurrent measured regime of dynamical stability.

The same reasoning requires the Alphonic Limit to be dynamical.

The smallest admissible distinction cannot be a static zero-duration entity. It must possess sufficient spatial and temporal persistence to participate in an interaction and generate registration.

The Alphonic Limit therefore becomes not the smallest thing, but the smallest admissibly registered dynamical distinction.

This preserves the central commitment of Finite Symbolic Mechanics:

physical claims begin at finite distinction.

Measured Dynamical Extent adds:

and finite distinction is itself dynamical.

The consequence is not yet a replacement physics.

It is a change in where physics begins.